

Program 7

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;

void code_function()
{
printf("Code address : %p\n", (void *)code_function);
}

int main()
{
static int static_var = 20;
int stack_var = 30;
int *heap_var;

heap_var = (int *)malloc(sizeof(int));
*heap_var = 40;

printf("PID = %d\n\n", getpid());

code_function();

printf("Global address : %p\n", (void *)&global_var);
printf("Static address : %p\n", (void *)&static_var);
printf("Heap address : %p\n", (void *)heap_var);
printf("Stack address : %p\n", (void *)&stack_var);

printf("\nProcess is running. Check /proc/%d/maps\n", getpid());

sleep(30);

free(heap_var);

return 0;
}
```

out put

```
collect2: error: ld returned 1 exit status
[ubuntu@ubuntu:~/OSLab/Practical$ gcc program7.c -o program7
[ubuntu@ubuntu:~/OSLab/Practical$ ./program7
PID = 10224

Code address : 0xc7b5d8c709a8
Global address : 0xc7b5d8c90010
Static address : 0xc7b5d8c90014
Heap address : 0xc7b5e7de9010
Stack address : 0xfffffc6abe69c

Process is running. Check /proc/10224/maps
[ubuntu@ubuntu:~/OSLab/Practical$
```